UNIVERSIDAD PRIVADA “FRANZ TAMAYO”

FACULTAD DE INGENIERÍA

SISTEMAS OPERATIVOS



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MATERIA: Sistemas Operativos

ACTIVIDAD: SERVIDOR DOCKER PERSONAL

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Paso 1: Verificación del entorno de la VM

Se ejecuta los siguientes comandos uno por uno para inspeccionar los recursos asignados:

hostname

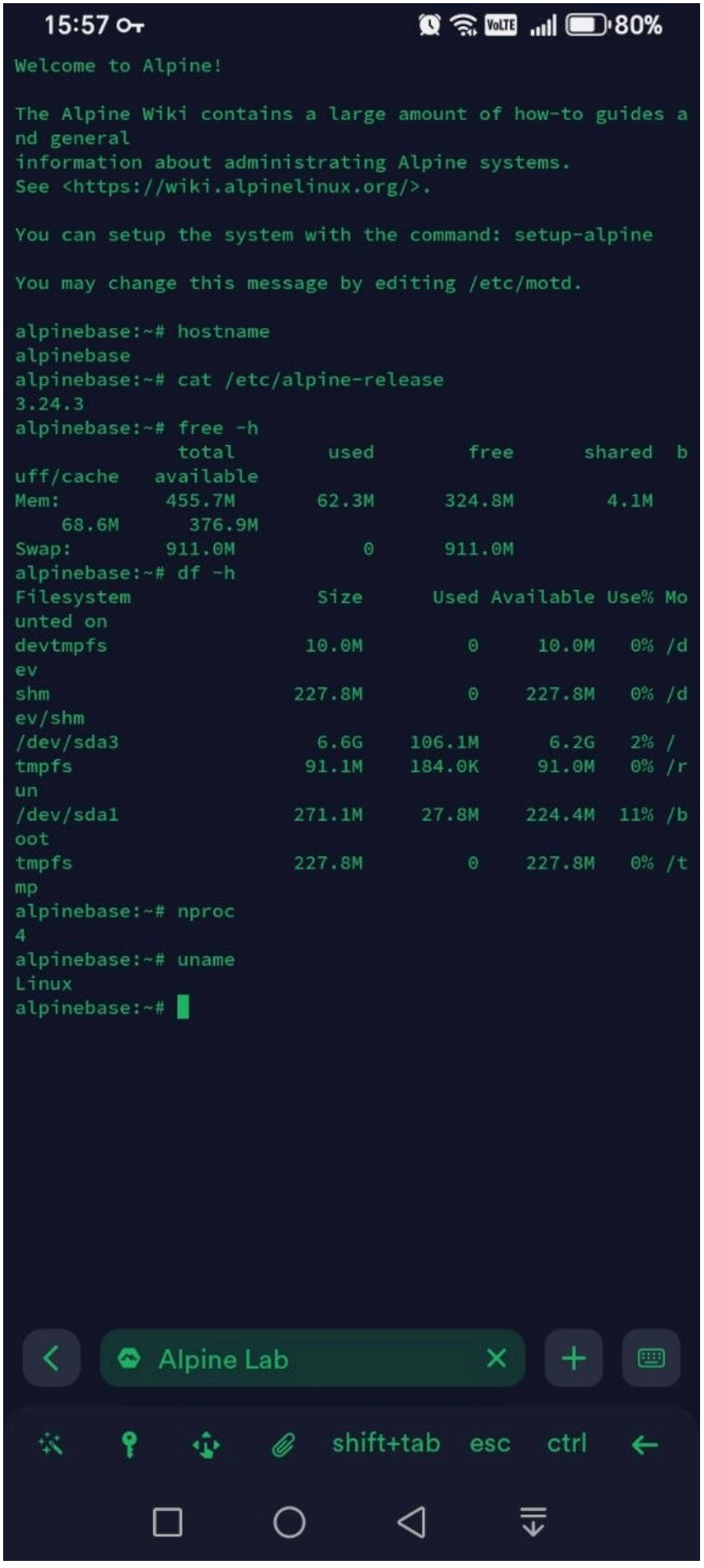
cat /etc/alpine-release

free -h

df -h /

nproc

uname -r


15:57
80%
Welcome to Alpine!
The Alpine Wiki contains a large amount of how-to guides and general
information about administrating Alpine systems.
See <https://wiki.alpinelinux.org/>.
You can setup the system with the command: setup-alpine
You may change this message by editing /etc/motd.
alpinebase:~# hostname
alpinebase
alpinebase:~# cat /etc/alpine-release
3.24.3
alpinebase:~# free -h
total used free shared buff/cache available
Mem: 455.7M 62.3M 324.8M 4.1M 68.6M 376.9M
Swap: 911.0M 0 911.0M
alpinebase:~# df -h
Filesystem Size Used Available Use% Mounted on
devtmpfs 10.0M 0 10.0M 0% /dev
shm 227.8M 0 227.8M 0% /dev/shm
/dev/sda3 6.6G 106.1M 6.2G 2% /
tmpfs 91.1M 184.0K 91.0M 0% /run
/dev/sda1 271.1M 27.8M 224.4M 11% /boot
tmpfs 227.8M 0 227.8M 0% /tmp
alpinebase:~# nproc
4
alpinebase:~# uname
Linux
alpinebase:~#
Alpine Lab
shift+tab
esc
ctrl

Paso 2: Instalación y arranque de Docker Engine

Se actualiza repositorios, instala Docker e inicia el servicio en OpenRC:

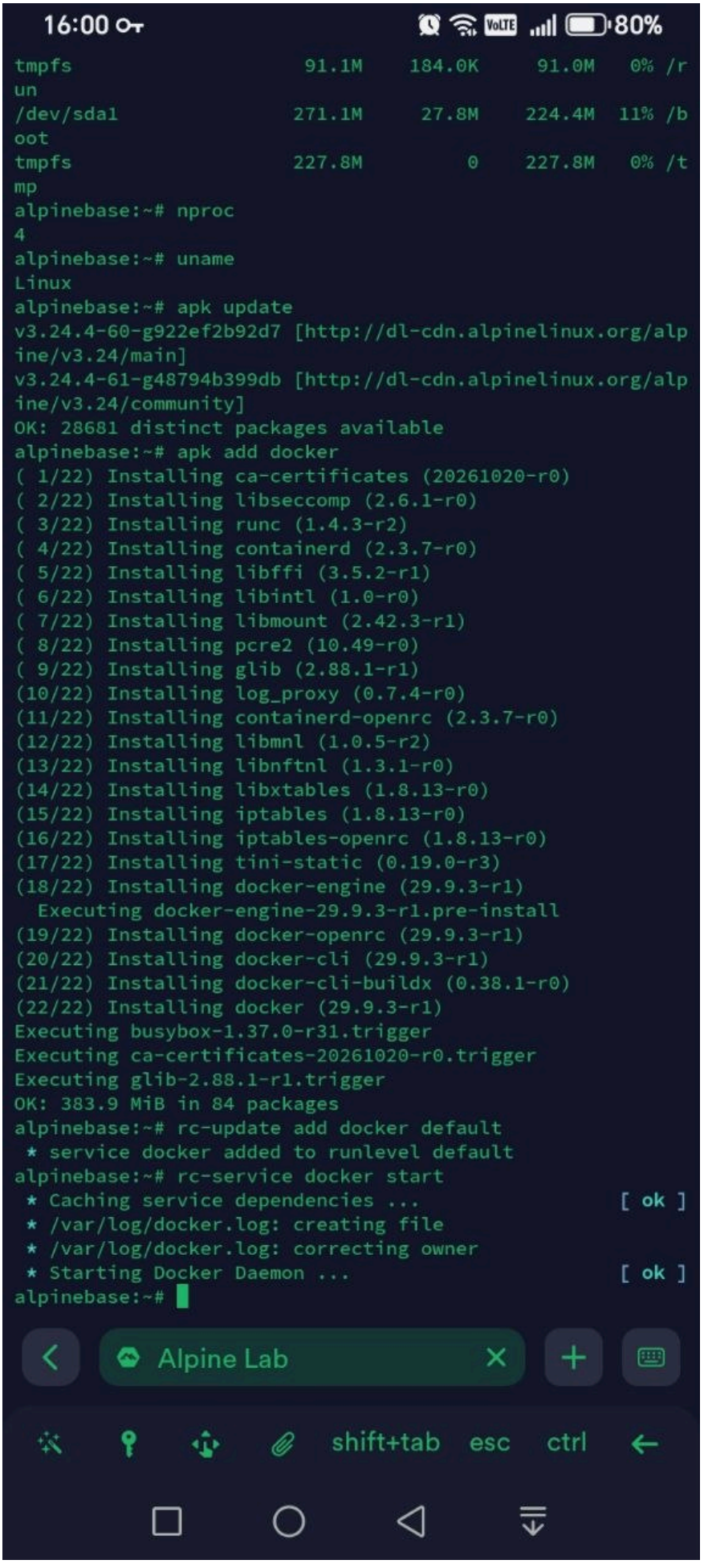
apk update

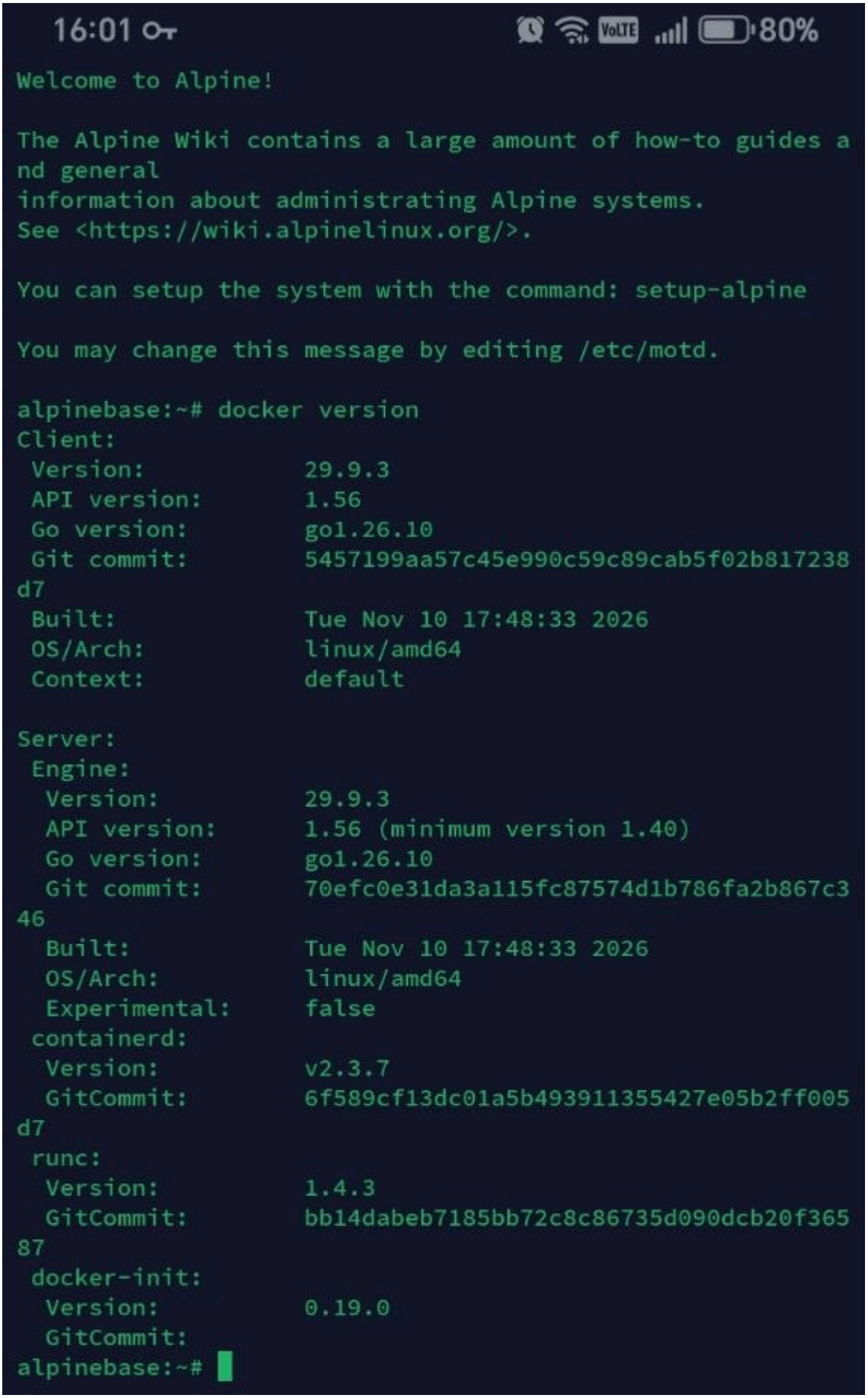
apk add docker

rc-update add docker default

rc-service docker start

docker version


16:00
80%
tmpfs 91.1M 184.0K 91.0M 0% /run
/dev/sda1 271.1M 27.8M 224.4M 11% /boot
tmpfs 227.8M 0 227.8M 0% /tmp
alpinebase:~# nproc
4
alpinebase:~# uname
Linux
alpinebase:~# apk update
v3.24.4-60-g922ef2b92d7 [http://dl-cdn.alpinelinux.org/alpine/v3.24/main]
v3.24.4-61-g48794b399db [http://dl-cdn.alpinelinux.org/alpine/v3.24/community]
OK: 28681 distinct packages available
alpinebase:~# apk add docker
(1/22) Installing ca-certificates (20261020-r0)
(2/22) Installing libseccomp (2.6.1-r0)
(3/22) Installing runc (1.4.3-r2)
(4/22) Installing containerd (2.3.7-r0)
(5/22) Installing libffi (3.5.2-r1)
(6/22) Installing libintl (1.0-r0)
(7/22) Installing libmount (2.42.3-r1)
(8/22) Installing pcre2 (10.49-r0)
(9/22) Installing glib (2.88.1-r1)
(10/22) Installing log_proxy (0.7.4-r0)
(11/22) Installing containerd-openrc (2.3.7-r0)
(12/22) Installing libmnl (1.0.5-r2)
(13/22) Installing libnftnl (1.3.1-r0)
(14/22) Installing libxtables (1.8.13-r0)
(15/22) Installing iptables (1.8.13-r0)
(16/22) Installing iptables-openrc (1.8.13-r0)
(17/22) Installing tini-static (0.19.0-r3)
(18/22) Installing docker-engine (29.9.3-r1)
Executing docker-engine-29.9.3-r1.pre-install
(19/22) Installing docker-openrc (29.9.3-r1)
(20/22) Installing docker-cli (29.9.3-r1)
(21/22) Installing docker-cli-buildx (0.38.1-r0)
(22/22) Installing docker (29.9.3-r1)
Executing busybox-1.37.0-r31.trigger
Executing ca-certificates-20261020-r0.trigger
Executing glib-2.88.1-r1.trigger
OK: 383.9 MiB in 84 packages
alpinebase:~# rc-update add docker default
* service docker added to runlevel default
alpinebase:~# rc-service docker start
* Caching service dependencies ... [ok]
* /var/log/docker.log: creating file
* /var/log/docker.log: correcting owner
* Starting Docker Daemon ... [ok]
alpinebase:~#
Alpine Lab
shift+tab
esc
ctrl


16:01
80%
Welcome to Alpine!
The Alpine Wiki contains a large amount of how-to guides and general
information about administrating Alpine systems.
See <https://wiki.alpinelinux.org/>.
You can setup the system with the command: setup-alpine
You may change this message by editing /etc/motd.
alpinebase:~# docker version
Client:
Version: 29.9.3
API version: 1.56
Go version: go1.26.10
Git commit: 5457199aa57c45e990c59c89cab5f02b817238d7
Built: Tue Nov 10 17:48:33 2026
OS/Arch: linux/amd64
Context: default
Server:
Engine:
Version: 29.9.3
API version: 1.56 (minimum version 1.40)
Go version: go1.26.10
Git commit: 70efc0e31da3a115fc87574d1b786fa2b867c346
Built: Tue Nov 10 17:48:33 2026
OS/Arch: linux/amd64
Experimental: false
containerd:
Version: v2.3.7
GitCommit: 6f589cf13dc01a5b493911355427e05b2ff005d7
runc:
Version: 1.4.3
GitCommit: bb14dabeb7185bb72c8c86735d090dcb20f36587
docker-init:
Version: 0.19.0
GitCommit:
alpinebase:~#

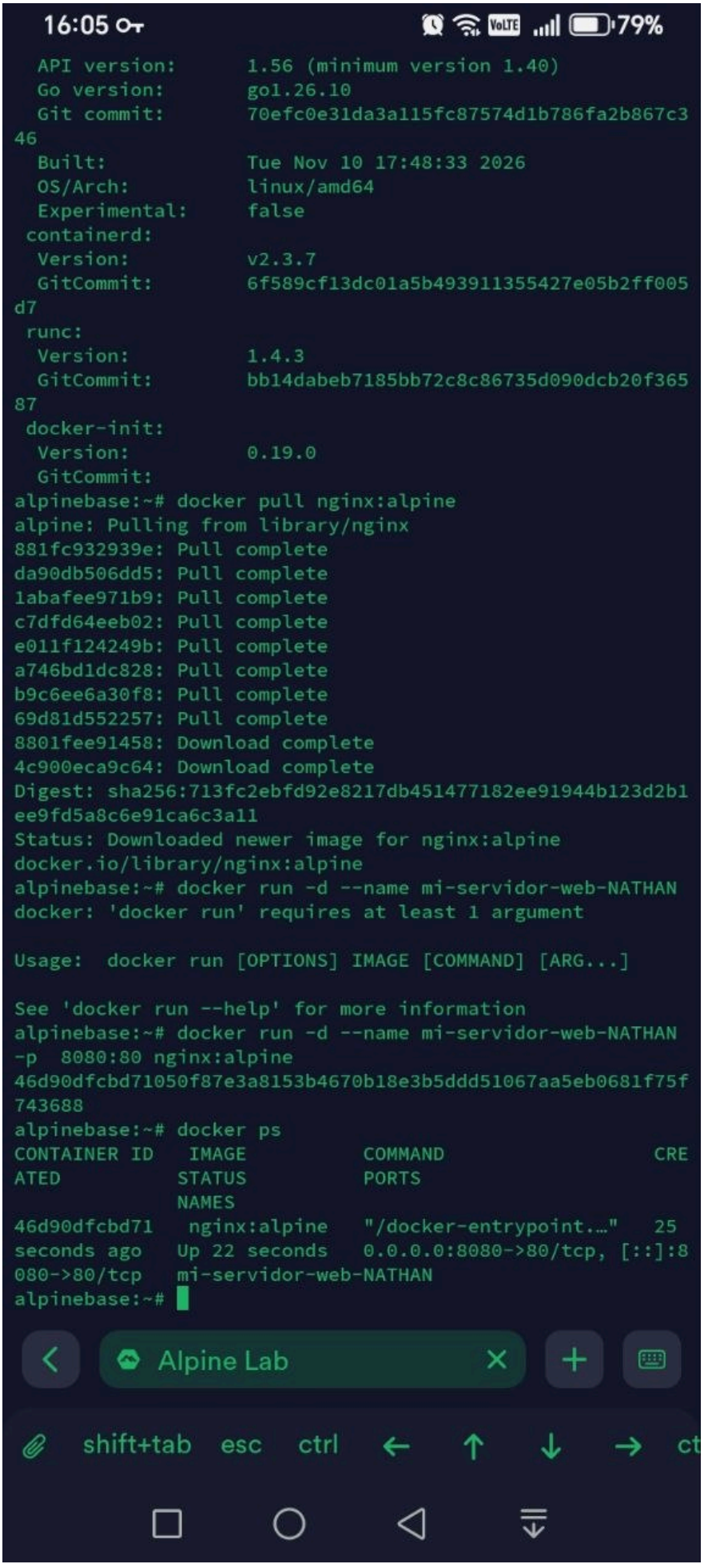
Paso 3: Despliegue del contenedor Nginx

Se descarga la imagen oficial de Nginx para Alpine y levanta el contenedor mapeando el puerto 8080 de tu VM al 80 del contenedor:

docker pull nginx:alpine

docker run -d --name mi-servidor-web -p 8080:80 nginx:alpine

docker ps


16:05
79%
API version: 1.56 (minimum version 1.40)
Go version: go1.26.10
Git commit: 70efc0e31da3a115fc87574d1b786fa2b867c346
Built: Tue Nov 10 17:48:33 2026
OS/Arch: linux/amd64
Experimental: false
containerd:
Version: v2.3.7
GitCommit: 6f589cf13dc01a5b493911355427e05b2ff005d7
runc:
Version: 1.4.3
GitCommit: bb14dabeb7185bb72c8c86735d090dcb20f36587
docker-init:
Version: 0.19.0
GitCommit:
alpinebase:~# docker pull nginx:alpine
alpine: Pulling from library/nginx
881fc932939e: Pull complete
da90db506dd5: Pull complete
1abafee971b9: Pull complete
c7dfd64eeb02: Pull complete
e011f124249b: Pull complete
a746bd1dc828: Pull complete
b9c6ee6a30f8: Pull complete
69d81d552257: Pull complete
8801fee91458: Download complete
4c900eca9c64: Download complete
Digest: sha256:713fc2ebfd92e8217db451477182ee91944b123d2b1ee9fd5a8c6e91ca6c3a11
Status: Downloaded newer image for nginx:alpine
docker.io/library/nginx:alpine
alpinebase:~# docker run -d --name mi-servidor-web-NATHAN
docker: 'docker run' requires at least 1 argument
Usage: docker run [OPTIONS] IMAGE [COMMAND] [ARG...]
See 'docker run --help' for more information
alpinebase:~# docker run -d --name mi-servidor-web-NATHAN -p 8080:80 nginx:alpine
46d90dfcbd71050f87e3a8153b4670b18e3b5ddd51067aa5eb0681f75f743688
alpinebase:~# docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
46d90dfcbd71 nginx:alpine "/docker-entrypoint.…" 25 seconds ago Up 22 seconds 0.0.0.0:8080->80/tcp, [::]:8080->80/tcp mi-servidor-web-NATHAN
alpinebase:~#
Alpine Lab
shift+tab esc ctrl

Paso 4: Creación y publicación del sitio web HTML

Se crea el archivo index.html dentro de tu VM copiando y pegando el siguiente bloque completo:

```
cat <<'EOF' > index.html

<!DOCTYPE html>

<html lang="es">

<head>

<meta charset="utf-8">

<meta name="viewport" content="width=device-width, initial-scale=1">

<title>Mi primera página con Docker</title>

<style>

*{box-sizing:border-box}

body{margin:0;background:#101d2d;color:#eef5fb;font-family:Arial,sans-serif}

header{padding:18px 74px;background:#14243a;border-bottom:1px solid #31506a}

header b{font-size:1.2rem;color:#5bd6d0}

main{max-width:1000px;margin:48px auto;padding:0 22px}

.tag{color:#65d6c2;font-weight:bold;letter-spacing:2px}

h1{font-size:clamp(2.3rem,6vw,4rem);margin:14px 0}
```

```
.lead{font-size:1.12rem;line-height:1.65;color:#c4d3df;max-width:680px}

.button{display:inline-block;margin-top:12px;padding:12px
20px;border-radius:30px;background:#087fc4;color:white;font-weight:bold}

.cards{display:grid;grid-template-columns:repeat(auto-fit,minmax(190px,1fr));gap:16px;mar
gin:34px 0}

.card{padding:22px;border-radius:15px;background:#1b3045;border:1px solid #31506a}

.card h2{color:#65d6c2;font-size:1.1rem}

footer{text-align:center;margin:40px 0 10px;color:#9db0bf}

</style>

</head>

<body>

<header><b>Mi Web con Docker</b></header>

<main>

<div class="tag">HITO 4 SISTEMAS OPERATIVOS</div>

<h1>¡Hola! Mi primera página web con Docker</h1>

<p class="lead">Este sitio se ejecuta dentro de un contenedor Docker utilizando la imagen
ligera de Nginx con Alpine Linux.</p>

<span class="button">Práctica del Hito 4</span>

<section class="cards">
```

<article class="card"><h2>Página personalizada</h2> Contenido creado por ti y publicado con Docker. </article>

<article class="card"><h2>Ligera y eficiente</h2> Nginx sirve esta página HTML sencilla. </article>

<article class="card"><h2>Objetivo</h2> Comprender cómo Docker ejecuta aplicaciones de manera aislada. </article>

</section>

<footer>Alpine Linux | Docker | Nginx</footer>

</main>

</body>

</html>

EOF

Además se verifica la creación del archivo y cópialo dentro del contenedor:

ls -lh index.html

docker cp index.html mi-servidor-web:/usr/share/nginx/html/index.html


16:31
75%
Mi primer...
47.21:8080
Mi Web con Docker
HITO 4 SISTEMAS OPERATIVOS
¡Hola! Mi primera página web con Docker
Este sitio se ejecuta dentro de un contenedor Docker utilizando la imagen ligera de Nginx con Alpine Linux.
Práctica del Hito 4
Página personalizada
Contenido creado por ti y publicado con Docker.